PROJECT REPORT

AI Equity Valuation and Portfolio Analytics

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Executive summary

The project examined how growth and reinvestment assumptions affect the value of AI-exposed companies, and whether an optimised allocation improved risk and return over equal weighting in a six-stock historical backtest.

Valuation. Oracle's revenue-target-aligned case produced an estimated enterprise value of **$483.78 billion**, compared with **$377.04 billion** under slower growth. The slower-growth estimate ranged from $303.17 billion to $506.97 billion across nine combinations of discount rate and terminal growth.

Portfolio. Equal weighting returned **109.77%** versus **80.94%** for the optimised allocation. It also had lower annualised volatility and a shallower maximum drawdown. Both portfolios began with $1 million and held the same six stocks.

What the project delivered

Stage	Deliverable
Python	DCF models, assumption sensitivities, and a historical comparison of two portfolios.
SQLite	Queryable valuation scenarios, performance results, and portfolio weights.
R	Independent annualised volatility calculation from exported daily balances.
Power BI	Two report pages covering performance, allocation, valuation, and sensitivity.

Enterprise value and sensitivity

The discounted cash flow models forecast revenue and after-tax operating profit, then subtract net reinvestment to estimate free cash flow to the firm. Enterprise value is the present value of those forecast cash flows and terminal value, discounted at the weighted average cost of capital (WACC).

Company	Scenario	Model baseline	EV (USD billions)
Alphabet	Illustrative	2025-12-31	2,012.84
Microsoft	Illustrative	2026-06-30	2,773.21
Oracle	Revenue target aligned	2027-05-31	483.78
Oracle	Slower growth	2027-05-31	377.04

Oracle's target-aligned case was **$106.74 billion higher**, or about 28.3% above slower growth. In that forecast, FY2030 revenue reaches $225.00 billion, while $118.44 billion of net reinvestment leaves free cash flow at **negative $55.83 billion**. Free cash flow turns positive in FY2031 as forecast reinvestment falls.

Oracle slower-growth case

Enterprise value in USD millions by WACC and terminal growth assumption.

WACC	2% growth	3% growth	4% growth
9%	424,090.44	458,622.01	506,966.21
10%	355,753.41	**377,040.25**	405,422.70
11%	303,171.59	316,426.59	333,468.73

At **10% WACC and 3% terminal growth**, $377,040.25 million reconciles to $377.04 billion in the scenario table. At 3% growth, moving WACC from 9% to 11% reduces the estimate by about $142.20 billion.

Microsoft reinvestment sensitivity. Sales-to-capital ratios of 0.5, 1.0, and 1.5 yielded estimates of $2,378.78 billion, $2,773.21 billion, and $2,904.68 billion, respectively.

Historical performance and risk

The two portfolios used AMZN, GOOGL, META, MSFT, NVDA, and ORCL. Equal Weight assigned approximately 16.67% to each stock. The Optimised portfolio had its largest allocations in Oracle, Alphabet, and Microsoft.

Test-period metric	Equal Weight	Optimised
Ending value	$2,097,708.80	$1,809,415.75
Cumulative return	109.77%	80.94%
Annualised volatility	27.16%	27.88%
Maximum drawdown	−27.04%	−28.05%
Concentration (sum of squared weights)	0.167	0.221
Effective holdings (inverse of concentration)	6.00	4.52

Equal Weight finished **$288,293.05 ahead**, a cumulative return gap of **28.83 percentage points**. It also recorded 0.72 percentage points less annualised volatility and a maximum drawdown 1.01 percentage points shallower. The Optimised portfolio's 4.52 effective holdings reflect the uneven distribution of capital across its six positions.

Validation checks

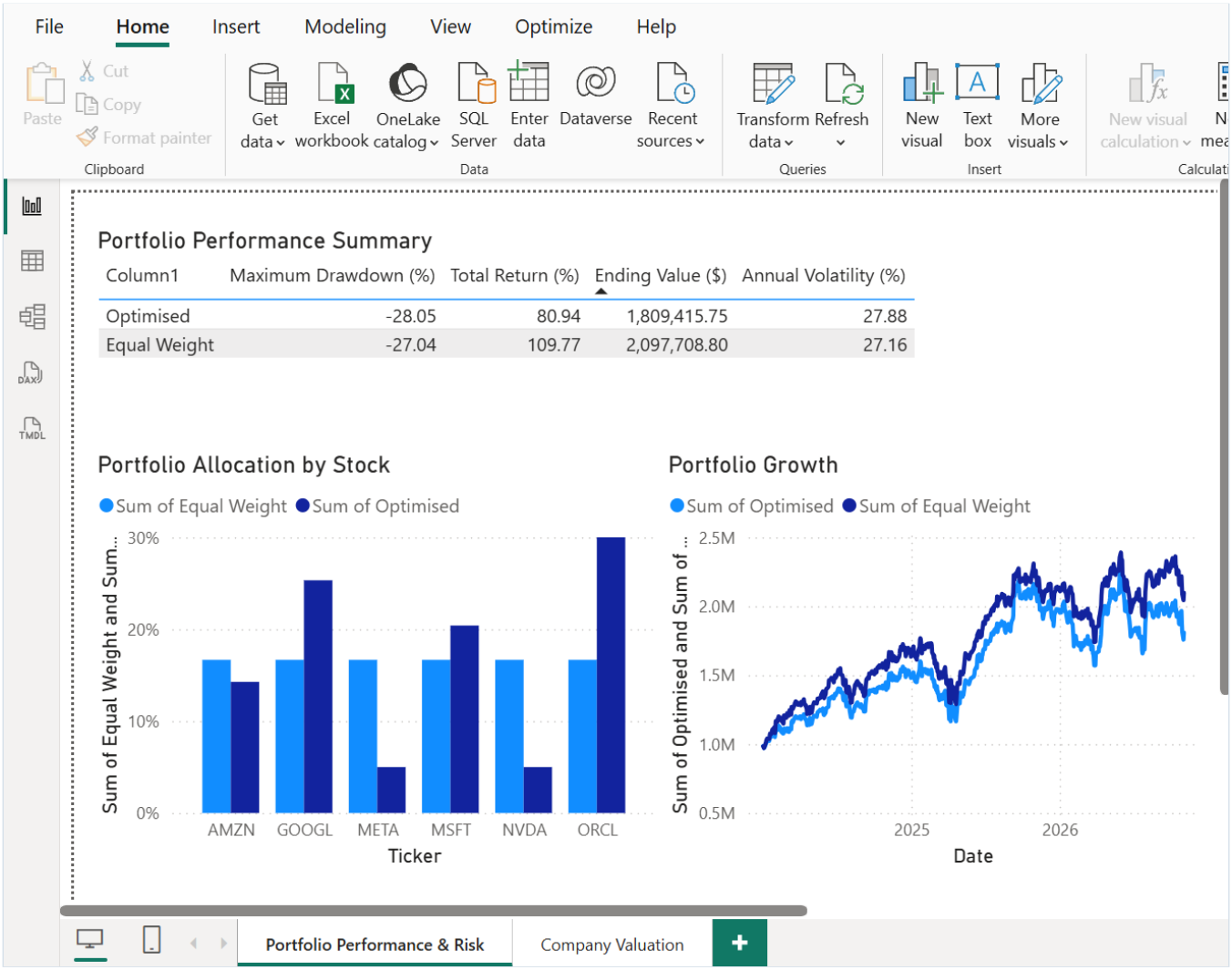
SQLite confirmed that both sets of weights summed to 100% and reproduced the concentration measures. In R, daily returns were reconstructed from 713 exported balances per portfolio and the $1 million starting value. Multiplying the sample standard deviation of daily returns by √252 reproduced the Python volatility results to two decimal places.

Scope of the findings

The backtest compares two allocations in one six-stock universe over one historical test period. The DCF results depend on the cash flow and discounting assumptions, and the companies use different model baseline dates. The Oracle sensitivity grid shows the range across the nine inputs tested.

Portfolio performance and risk

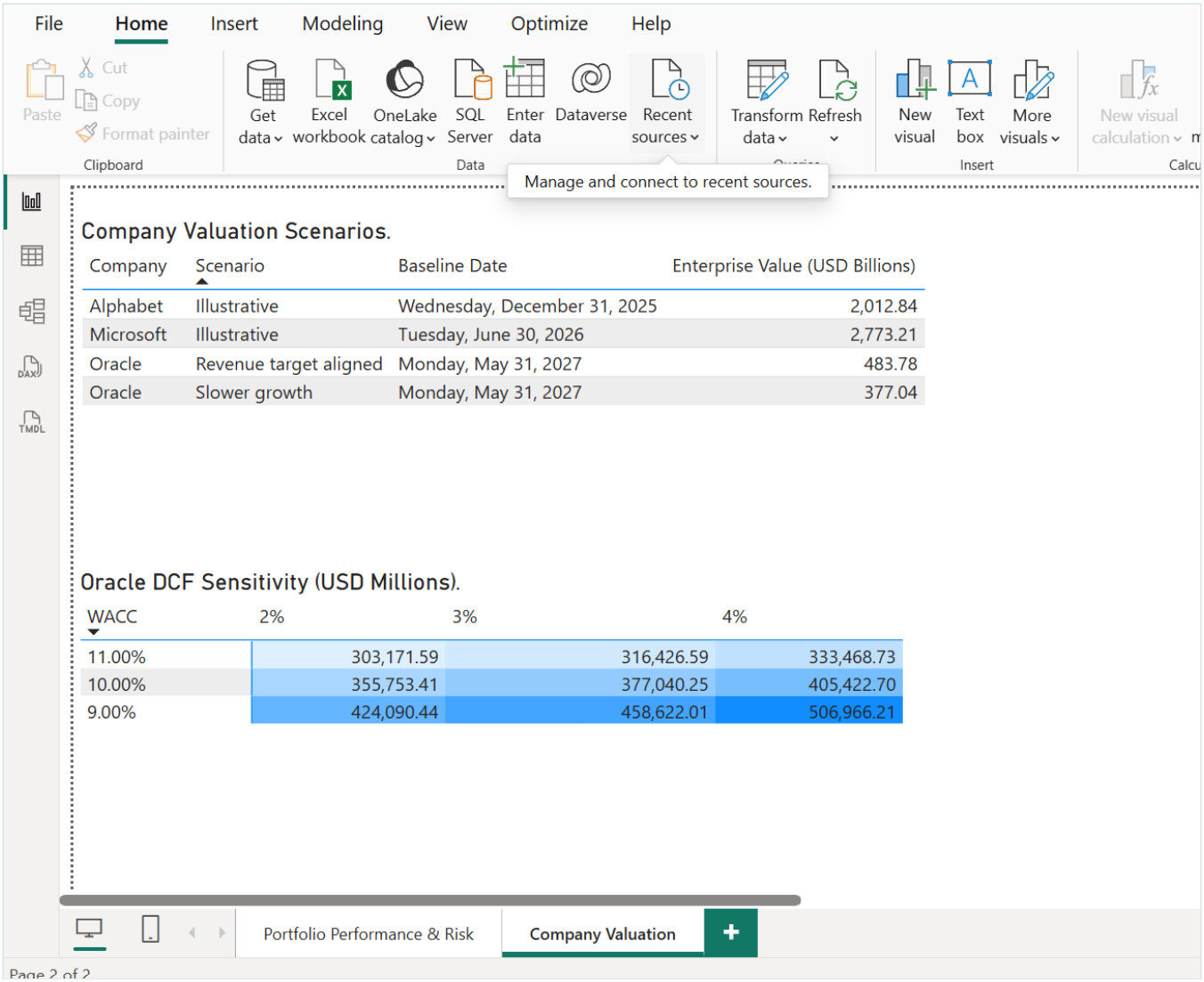
The report page combines the test-period summary, stock-level weights, and daily portfolio values.



Source: Portfolio Performance & Risk page from the project's Power BI report.

Company valuation

The company table reports enterprise value in USD billions. The Oracle heatmap reports enterprise value in USD millions.



Source: Company Valuation page from the project's Power BI report.